Wildfire Risk Analytics and Prediction

Data Analytics / Machine Learning Internship Project

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Dataset: Synthetic demonstration dataset created for this project

1. Abstract

This project demonstrates an end-to-end data analytics workflow for identifying areas with higher wildfire risk. The workflow covers data preparation, exploratory data analysis (EDA), predictive modeling using Random Forest, feature-importance analysis, and unsupervised clustering. The dataset is synthetic and is intended only for academic demonstration; the results must not be interpreted as operational wildfire forecasts.

2. Problem Statement

Wildfire response planning can benefit from structured analysis of environmental and location-related indicators. The objective here is to build a reproducible analytical pipeline that classifies areas into a binary wildfire-risk outcome and groups areas with similar characteristics.

3. Dataset and Features

Feature	Description
temperature_c	Temperature in °C
humidity_pct	Relative humidity (%)
wind_speed_kmh	Wind speed (km/h)
rainfall_mm	Rainfall (mm)
vegetation_dryness	Synthetic dryness index from 0 to 1
population_density	Synthetic population density indicator
distance_to_road_km	Distance to nearest road (km)
elevation_m	Elevation (m)
wildfire_risk	Binary synthetic target: 0/1

4. Methodology

• Data exploration: descriptive statistics, missing-value checks, regional risk summaries, and correlations.

• Data preparation: selection of numerical predictors and an 80/20 stratified train-test split.

• Predictive modeling: Random Forest classifier with class balancing.

• Model evaluation: accuracy, classification report, and confusion matrix.

• Interpretability: Random Forest feature importance.

• Unsupervised learning: K-Means clustering after standardization.

5. Results

The model achieved an accuracy of approximately **70.83%** on the held-out synthetic test set. Because the data-generating process is synthetic, this accuracy is not evidence of real-world predictive performance.

Metric	Value
Rows	1,200
Training rows	960
Test rows	240
Test accuracy	70.83%
Number of clusters	4

6. Feature Importance

Rank	Feature	Importance
1	vegetation_dryness	0.2101
2	temperature_c	0.1590
3	humidity_pct	0.1465
4	wind_speed_kmh	0.1187
5	rainfall_mm	0.1003
6	distance_to_road_km	0.0930
7	population_density	0.0921
8	elevation_m	0.0802

7. Key Findings

• The synthetic dataset contains 1,200 area records across five regions.

• The Random Forest model provides a reproducible baseline for binary risk classification.

• Feature importance identifies which synthetic predictors contributed most to model decisions.

• K-Means creates four groups of areas with similar environmental/location profiles.

• The project demonstrates the complete analytics lifecycle from data preparation through presentation-ready outputs.

8. Limitations and Ethical Considerations

The dataset is synthetic. It does not represent actual wildfire incidents, locations, weather observations, or emergency-service capacity. Therefore, the model should not be used for real firefighter deployment or public-safety decisions. A real deployment would require validated historical wildfire records, weather data, geospatial information, careful bias analysis, external validation, monitoring, and domain-expert review.

9. Future Scope

• Integrate validated historical wildfire and meteorological datasets.

• Add geospatial maps and time-series forecasting.

• Compare multiple models such as Logistic Regression, Gradient Boosting, and calibrated ensembles.

• Build an interactive Power BI/Tableau dashboard.

• Use cross-validation and external validation before considering operational use.

10. Conclusion

The project provides a compact demonstration of data cleaning, EDA, predictive analytics, machine learning, visualization, and clustering in one reproducible workflow. It is suitable as an academic portfolio project and can be extended when an appropriate real dataset is available.